



Automatic Helmet Detection from Videos Using Artificial Intelligence Method and Engage with Secured Driving System by Message Alert

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Received: 10 July 2025 / Accepted: 2 November 2025

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Abstract

Motorcycle usage is growing globally and one of the main reasons is affordability. Despite helmet laws, many riders don't wear them. Not wearing helmets may enhance the threat of brutal head trauma, death and coma. Researchers have applied surveillance video systems to detect traffic violations, particularly for riders without helmets. While deep learning algorithms for helmet detection in road surveillance have been developed, less focus has been given on adapting them for the complex environments of construction sites. Most of the traditional methods of helmet detection rely on image processing or manual inspections, which can be time-consuming and inaccurate. Hence, an automatic motorcycle and helmet detection mechanism is implemented in this paper with the help of deep learning strategies. At first, the requisite videos are fetched from the available data sources. From the fetched videos, the motorcycle is detected with support of Adaptive Yolov5 network. Here, the network performance is improved by optimizing the parameters using Improved Hermit Crab Optimizer (IHCO). Subsequently, the Region of Interest (ROI) is extracted from the detected video frames. Furthermore, the Vision Transformer-based Dilated Residual DenseNet (ViT-DRDNet) is introduced to recognize the helmet from the ROI extracted frames. If there is no helmet detected in the specific motorcycle, then the number plate of the motorcycle is recognized by Adaptive Yolov5 network. Based on the detected number plate of the vehicle, the message alert is sent to the registered phone number, which facilitates the secured driving system. Lastly, the performance and supremacy of developed motorcycle and helmet detection mechanism are estimated among various existing techniques and algorithms. The accuracy score of the proposed helmet detection mechanism is 94.95, showing its overall ability in enhancing the road safety of the public with accurate detection outcomes.

Keywords Automatic helmet detection · Secured driving system · Message alert · Adaptive Yolov5 · Improved hermit crab optimizer · Vision transformer-based dilated residual DenseNet

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1 Introduction

Based on World Health Organization (WHO), more than 1.3 million citizens lose their lives annually due to accidents (Li et al. 2022). The accidents occur because of poor driving strategies as well as human error. Motorcycles are the major cause of such accidents, and it is an important source of serious damage in small countries (Hao et al. 2023). However, helmets help to reduce the risk of head injuries and prevent fatalities among the riders of motor vehicles. Helmet-wearing enforcement becomes a significant strategy for reducing mortality and fatalities. Many nations depend on traffic administrators to ensure compliance with traffic regulations by openly examining the drivers regularly (Siebert 2020). However, this approach generally incurs high costs and often complex to organize many traffic officers in each country to guarantee strict and universal enforcement (Nandhini and Brindha 2023). Moreover, computer-based applications play vital role in monitoring more challenging behavior of road users, such as using mobile phones while driving and illegally using carpool lanes (Li et al. 2024).

From the beginning, the method based on removing the background has employed for extracting moving vehicles and objects from the collected videos. Then, deep learning and machine learning models are applied to determine the motorcycles (Li et al. 2023). Moreover, the motorcyclist's head area is identified, and another model is then employed to classify the usage of the helmet from non-helmet use (Song et al. 2023). The handcrafted features are utilized to enhance the classifier's performance, and most common model is Histogram-Oriented Gradient (HOG), which helps to retrieve the deep features present in the identified head regions (Chen 2023). However, such models cannot provide efficient performance if there are too many motor vehicles and motorcyclists. Instead of implementing handcrafted features, the methods-based deep learning can autonomously develop representations from source image information that are most appropriate for detecting the task of helmet usage (Li 2023). While recorded video data with traffic monitoring infrastructures are intrinsically based on frames, helmet usage data generated through automated detection should be applied to every vehicle to allow effective assessment of helmet usage (Zhao et al. 2023; Xu et al. 2023). Several existing approaches lack this surveillance operation and have several limitations (Shan 2023). The existing techniques apply the attention network for spatial or channel data extraction, which leads to remote dependence and lack of global consideration of all information (Zeng et al. 2023).

Computer vision aims to give machines the capacity to see and comprehend visual data from their surroundings which creates opportunities for usage in augmented reality, video processing, and image analysis. The YOLO algorithm

is a ground-breaking method for object recognition in the field of computer vision (Prakash-Borah 2024). In recent times, researchers have developed several neural network models to identify the use of safety helmets (Song and Wang 2023). Many previous articles have provided insight into this issue (Huang et al. 2021). Numerous techniques have been developed based on Convolutional Neural Networks (CNNs) with varying depths for detecting helmet usage (Jia et al. 2021). However, such models present some drawbacks in pre-processing like image sharpness, object quantity, and color contrast between the foreground and background (Huang et al. 2021; Siebert et al. 2024). To conquer all these issues, the automated helmet detection model is developed to employ a deep learning strategy.

Scope of the Research The scope of this research is explicitly centered on enhancing road safety monitoring through automatic helmet detection in motorcycle riders, rather than extending to construction-site or industrial safety scenarios. The study leverages video streams from roadside surveillance systems to detect motorcycles, identify whether riders are wearing helmets, and, in the case of violations, automatically recognize the vehicle's number plate and generate secured message alerts to registered user. By incorporating the Adaptive YOLOv5 with parameter optimization by IHCO and integrating the ViT-DRDNet, the system is designed to operate effectively in real-world road conditions, where challenges such as traffic congestion, diverse lighting, varying riding postures, and occlusions frequently occur. Importantly, the research does not encompass helmet detection in non-traffic environments, such as construction sites or industrial zones, which require different environmental modeling and safety compliance mechanisms. This distinction ensures that the scope is well-defined, application-specific, and directly aligned with overarching goal of reducing motorcycle-related fatalities through reliable and automated helmet compliance monitoring.

The contributions of the developed model are given below:

- To implement the A-YOLOV5-based motorcycle detection model that aids in effectively detecting the motorcycle to improve security by enforcing rules in various places. It also supports to recognize the number plate when no helmets are detected. The Adaptive YOLOv5 network, further optimized by IHCO, is used to accurately recognize the vehicle's number plate.
- To optimize the performance of the motorcycle detection system by employing the IHCO that helps to fine-tune the parameters of Adaptive YOLOv5 network. The optimization improves detection accuracy, reducing

false positive results that are helpful for real-time traffic monitoring systems.

- To suggest the ViT-DRDNet-based helmet detection model assists in detecting the helmet to promote an ethic of security and lower the likelihood of accidents. This model captures long-range dependencies in the motorcycle-detected frames while maintaining fine-grained details, making it highly effective for detecting helmets under various occlusion, lighting, and angle conditions.

The workflow of the suggested helmet detection model is as follows. Automatic helmet detection framework from videos for secured driving systems by message alert using artificial intelligence method is depicted. Detecting the motorcycle from videos through the detection process is provided. The deep learning model for helmet detection and adaptive model number plate recognition for the secured driving system by message alert are shown. The result and discussion are elaborated.

2 Literature Survey

This section reviews relevant studies in the domain of helmet and vehicle detection systems.

2.1 Related Works

A discussion of previous works in helmet detection and motorcycle recognition systems are given in the following:

In 2020, Lin et al. (2020) have proposed an effective CNN model of Multi-Task Learning (MTL) to identify and track individual motor vehicles. In 2020, Li et al. (2020) have recommended a model using deep learning for the immediate identification of the safety helmet on the building site. In 2021, Han et al. (2021) have proposed a new object recognition model using a Single Shot Detector (SSD) to resolve the difficulty of less truthfulness in conventional techniques. In 2023, Bouhayane et al. (2023) have addressed the complex issue of automatic surveillance in the usage of helmets through machine learning and computer vision. In 2022, Sadiq et al. (2022) have developed efficient solutions for helmet safety identification issues. In 2022, Chen et al. (2022) have proposed an aerial remote sensing identification paradigm for cyclists' helmet-tiring detection. In 2023, Lee et al. (2023) have separated the data provided as input into the hat, helmet, and head categories, and trained the model to identify hats worn by workers. In 2023, Dasgupta et al. (2023) suggested a system to identify helmetless motorbikers. In 2024, Kumar and Asif (2024) have recommended a model for identifying the motorcyclists, who were riding without helmets. In 2024, Borah et

al. (Prakash-Borah 2024) have suggested a model to check whether the motorcycle riders were wearing a helmet or not. In 2024, Sarmah et al. (2024) have adopted a yolov5 mechanism for recognizing the helmets of motorcycle riders to ensure that all the bike riders were following the safety rules. In 2024, Alassaf and Said (2024) have designed the Deformable Perspective Perception Network (DPPNet) techniques for overcoming issues like occluded helmet detection and small object miss-detection.

2.2 Problem Statement

The video surveillance-aided automated identification of the helmet has the potential to enhance road safety. Some of the limitations of conventional methods are given below:

- Conventional motorcycle and helmet detection models do not have the capacity to handle complicated and dynamic video situations like changing lighting, occlusions, and diverse angles those results in low accuracy. These models may miss detecting motorcycles or helmets in certain scenarios, leading to unreliable security monitoring.
- When it comes to motorbike and helmet identification, many of the existing detection systems struggle to optimize the network parameters, which might lead to less ideal performance. Such algorithms could miss important things in video footage when sophisticated optimization techniques aren't used, which could compromise security and result in high false-negative rates.
- Traditional techniques frequently use generic algorithms that don't concentrate on ROI extraction, which can lead to slower detection and wasted processing resources. The efficacy of the detection procedure is diminished by inefficiency, particularly in real-time applications.
- Automatic alarm mechanisms based on vehicle identification are not frequently integrated with existing systems, particularly when a helmet is not identified. Thus, efficient road security is not guaranteed in crucial circumstances that lead to unsafe driving that compromises overall security.

Thus, an automated helmet detection approach has been implemented with the support of deep learning techniques. Various helmet-detection approaches have been introduced to ensure road safety.

3 Automatic Helmet Detection Framework for Secured Driving System

This section explains the proposed framework for automatic helmet detection using video inputs, which forms the foundation of a secured driving system with real-time message alerts.

3.1 Proposed Helmet Detection Framework

This research introduces an automated system designed to enhance motorcycle safety through real-time helmet and number plate detection. The process begins with collecting videos from standard data sources. Further, the A-Yolov5-based network is employed to accurately detect motorcycles, demonstrating robust performance even in complex environments. Here, the network performance is improved by optimizing the network parameters like step per epoch, hidden neurons, and number of epochs with the assistance of IHCO to enhance the IoU and accuracy. IHCO helps to fine-tuning the network parameters, resulting in faster and more efficient detection with high accuracy. Subsequently, the ROI is extracted from the detected video frames. By extracting the ROI, the processing speed is improved and computational load is minimized, which helps to focus on the relevant parts of the video frames. Furthermore, the ViT-DRDNet is introduced to detect the helmet from the ROI-extracted frames. Even intricate details are precisely analyzed by ViT-DRDNet, which aims to minimize the false positive outcomes during helmet detection. If there is no helmet detected in the particular motorcycle, then the number plate of the motorcycle is recognized by the A-Yolov5 network. This method helps to ensure that violators are identified based on their vehicle's registration, making the work easier for higher officials to assign fines to that person for violating the rules. Based on the detected number plate of the vehicle, the message alert is sent to the registered phone number, which facilitates the secured driving system. Figure 1 illustrates the pictorial depiction of the developed model.

3.2 Description of Two Wheeler Driving Dataset

The video frames needed to perform the process of the developed helmet detection model are as follows.

YouTube-1: The video frames are taken using the link <https://youtu.be/pZJO7MPIN0U?si=rerI3XpATKg-aIjZ>. Access date: 2024-03-09. The data source for this experiment is created by employing a single publicly available YouTube video named “Latur Old Collector Office Wrong side Romeos”, which shows the real-world traffic violations near the Old Collector Office in Latur, Maharashtra.

It contains continuous roadside surveillance footage for capturing the riders and motorcycles under natural conditions. It helps to identify the people who are not wearing the helmets. By using this video, the developed model collects several images for the helmet detection process. The total duration of the video is 18.51 min.

- **Source:** YouTube (link: YouTube Search).
- **Duration:** The full duration of the video is processed, segmenting it into individual frames for validation.
- **Frame Rate:** Frames are sampled at regular intervals to guarantee meaningful coverage of distinct events.

Also, the video is recorded in natural outdoor settings during the daytime, thus providing the consistent lighting. The distribution is explained below.

Lighting conditions:

- **Daylight:** 100% of the video is recorded under clear daylight conditions. No nighttime, artificial lighting or low-light scenes are present.
- **Shadows:** Several portions of the footage contain partial shadow because of trees and buildings, allowing moderate testing of the model's robustness under uneven lighting.

Weather conditions:

- **Clear weather:** The overall video is captured on clear data with no fog, rainfall, or adverse weather.
- **Temperature and humidity:** These factors are not explicitly estimated, yet typical daytime weather in Latur suggests warm and dry conditions.

Each frame in the video includes 4 to 8 motorcyclists. However, accurately calculating the motorcyclist's count is difficult in this video as it varies in each frame. Also, only, 20% of people are wearing helmets in this video.

YouTube-2: The video frames are taken from <https://www.youtube.com/watch?v=U-VXhkWw1P8>. It has diverse traffic scenarios captured across urban, rural, and semi-urban regions, and includes significant variations in helmet color, size, and design, as well as differences in lighting conditions, weather effects, and camera angles. In a scenario, where video data is processed for tasks like motorcycle and helmet detection, the specified parameters of RGB color encoding like 512×512 frame size, and 15,000 frames within a 5-minute duration are involved in this link.

Thus, the developed work utilizes a publicly available YouTube video for validation. This video is converted to 3399 frames for the designed approach, where 75% frames are selected for training and 25% frames are selected for

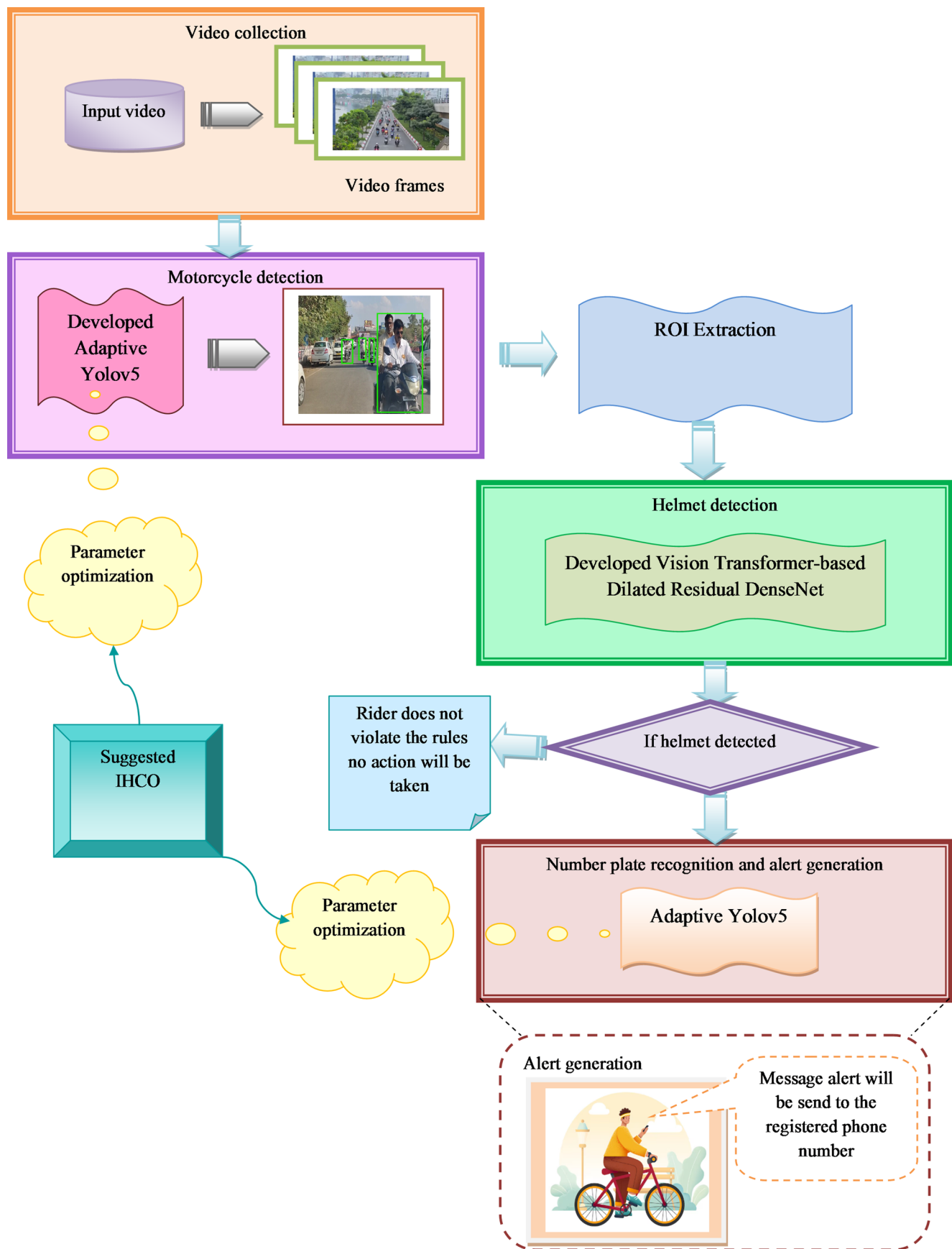


Fig. 1 Pictorial depiction of the proposed automatic helmet detection model

testing purposes. The collected video frames are defined using the term Cm_B^{MC} and the total counts of frames are signified as B . The term MC describes the traffic videos.

4 Motorcycle Detection Using Adaptive Yolov5 with Optimized Parameters

This section discusses the methodology for motorcycle detection from input video using the suggested Adaptive Yolov5 model.

4.1 Implemented IHCO-aided Parameter Tuning

The developed IHCO heuristic approach is the improvised version of the HCOA. Backpropagation algorithm-based optimization works well for deep learning model training, but it has issues like complex optimization landscapes and local minima. In this work, the model's hyper-parameters are optimized using the IHCO, which helps to avoid these problems and explores a wider solution space.

Reason for Choosing the HCOA: Although backpropagation works well for fine-tuning weights in deep learning models, it is very dependent on gradient information and may not be able to optimize in complex problem spaces or non-convex landscapes. The Particle Swarm Optimization (PSO) algorithm has better global search ability, yet it has local minimum problems and the popular Artificial Bee Colony (ABC) algorithm does not provide efficient solutions in large-scale problems. However, population-based evolutionary algorithms such as HCOA are able to investigate the solution space more widely, which reduces issues like local optima and offers potential solutions. The HCOA (Tafakkori and Tavakkoli-Moghaddam 2022) algorithm has several significant qualities that are: it was motivated by hermit crabs' capacity to quickly adjust regardless of circumstances. This approach works well at dealing with dynamic optimization issues in which the circumstances shift over time. However, as with many optimization techniques, the HCOA can prove computationally costly, particularly for massive problems. Because of the trade-offs associated with the advantages and disadvantages of the HCOA, we modified it and suggested an effective IHCO.

Novelty The IHCO introduces significant advancements over traditional optimization techniques in the context of deep learning-based motorcycle detection. Unlike standard optimization, which relies heavily on gradient information and often struggles with non-convex or complex optimization landscapes, IHCO explores a broader solution space, thereby reducing the risk of being trapped in local minima. The algorithm enhances the original HCOA by

incorporating mechanisms that dynamically adjust the random components used in updating the position, as shown in Eq. (1), which allows for more effective exploration and exploitation of the search space.

$$\alpha = \frac{\left(\frac{Q_{Y_L}}{F_{M_W}}\right)}{\left(\frac{N_{R_F}}{F_{M_W}}\right)} \quad (1)$$

Here, the best-fit, mean-fit and worst-fit values are defined utilizing the terms Q_{Y_L} , F_{M_W} and N_{R_F} . Updating the random value balances the trade-off between exploration and exploitation better than other population-based algorithms like PSO or ABC, which either risk premature convergence or fail to efficiently handle large-scale optimization problems. IHCO systematically updates the positions of candidate solutions based on best-fit, mean-fit, and worst-fit values. This structured parameter tuning ensures that the YOLOv5 network achieves faster convergence, improved generalization, and superior detection performance, making the IHCO-aided optimization a novel and critical contribution to the proposed motorcycle and helmet detection framework. In the proposed framework, IHCO is specifically applied to optimize key parameters of the YOLOv5-based motorcycle detection model, including the number of hidden neurons, steps per epoch, and the total number of epochs. This targeted optimization directly improves critical performance metrics, such as IoU and detection accuracy, by ensuring that the model is trained under near-optimal configurations. The adaptive search strategy of IHCO allows the model to handle dynamic and complex video environments, including varying lighting, motion blur, and occlusions, by fine-tuning parameters that significantly influence learning efficiency and convergence speed.

Role of the Optimization Algorithm In the developed ViT-DRDNet for helmet detection, the IHCO optimization algorithm plays a crucial role in fine-tuning the model's hyper-parameters, particularly the number of hidden neurons, the number of epochs, and the steps per epoch. These parameters significantly influence the learning dynamics and overall detection accuracy of the model. The number of hidden neurons determines the network's capacity to capture complex feature representations from visual data. Similarly, the number of epochs governs how long the model trains, where insufficient training leads to poor generalization. The steps per epoch dictate the granularity of weight updates within each epoch, directly affecting convergence stability and learning efficiency. By leveraging IHCO, the optimization process systematically explores a wide range of candidate configurations using best-fit, mean-fit, and worst-fit solutions, which ensures that the trade-off between exploration and exploitation is effectively managed. This

enables the model to avoid local minima, achieve faster convergence, and improve robustness under varying conditions such as occlusion, motion blur, and illumination changes. Algorithm-1 illustrates the pseudo-code of the introduced IHCO algorithm.

Algorithm 1: Developed IHCO

Input: Initial parameters
Initialization of population and parameters
While ($m < M$) do
Amend the random value α using the adaptive concept in Eq. (1)
Determine the availability of shell
If $fit < F$ then
Determine the position of the best crab
Update the location of odor
Else If
Perform the shell exchanging process
Find the local optimal solution
End if
End while
Return best solution
Output: Optimized hidden neurons $Y_{c_i}^{YOVS}$, epochs Ab_d^{YOVS} and step per epoch $O_{t_b}^{YOVS}$

4.2 Yolov5

YOLOv5 (Jia 2021) outperforms all earlier iterations in terms of accuracy as well as speed. The YOLOv5 algorithm generates four different variants of the approach, called (i) YOLOv5s, (ii) YOLOv5m, (iii) YOLOv5l, and (iv) YOLOv5x, by varying the depth and width of the backbone system using the variables depths-multiple and widths-multiple. The most basic variant, YOLOv5s, has the shortest parameter set with the quickest recognition rate. Four horizontal sliced layers make up the majority of the block focus, which deals with the incoming image.

4.3 Developed A-Yolov5-based Motorcycle Detection Model

In this study, an A-YOLOv5 network is developed specifically for motorcycle detection. The model not only detects motorcycles efficiently in complex and dynamic environments but also integrates an optimization mechanism using IHCO to fine-tune the network parameters, thereby it enhances detection performance under varying conditions. This approach ensures that motorcycles are reliably detected in diverse scenarios, forming a strong foundation for subsequent helmet recognition and automated safety interventions.

Novelty In the proposed framework, the collected video frames Cm_B^{MC} are first inputted into the YOLOv5 model, which is specifically designed to detect motorcycles in

real-time video streams. The image frames are fed into the YOLOv5 model, which performs feature extraction using its backbone network to identify key motorcycle characteristics. YOLOv5 has emerged as one of the most effective deep learning frameworks for real-time object detection due to its high accuracy and computational efficiency. In the context of motorcycle detection, YOLOv5 contains several advantages over traditional detection methods. Its single-stage detection architecture allows the model to predict object locations and classes simultaneously, significantly reducing inference time compared to multi-stage detectors. The network generates bounding box predictions and confidence scores for each detected object, followed by non-maximum suppression to eliminate overlapping detections. YOLOv5's multi-scale detection layers ensure motorcycles of different sizes and distances are accurately recognized. To further improve performance, adaptive anchor box generation and parameter optimization can be applied, allowing the model to adjust dynamically to variations in motorcycle shapes, occlusion, and environmental conditions. Unlike conventional YOLOv5, the adaptive version integrates advanced parameter optimization, enhancing both accuracy and IoU performance. Critical parameters, including hidden neurons, the number of training epochs, and steps per epoch, are dynamically optimized using the IHCO, allowing the model to adapt to diverse datasets and environmental conditions. An adaptive version of YOLOv5 helps to maintain a better balance between detection accuracy and processing speed. This makes it especially appropriate for security or surveillance systems that rely on video, where processing in real-time is essential. Finally, the detected motorcycles are tracked across frames, enabling real-time monitoring and downstream applications such as helmet detection or automated alert systems.

The developed network is capable of learning robust feature representations of motorcycles, which vary in shape, size, and orientation across different traffic and environmental conditions. Additionally, YOLOv5 supports multi-scale detection, enabling it to detect motorcycles at various distances from the camera, including small or partially occluded vehicles. The modular design allows for easy integration of improvements, such as adaptive anchor box generation and parameter optimization, which further enhance detection performance. By processing video frames in real-time, YOLOv5-based motorcycle detection systems can provide instantaneous alerts, making them suitable for traffic monitoring, helmet compliance enforcement, and intelligent surveillance applications. Furthermore, the framework's compatibility with data augmentation techniques improves its generalization ability, ensuring accurate detection even

when trained on limited datasets or in complex scenarios with motion blur, lighting variations, and occlusions.

Equation (2) shows the objective function CM_{CCY} for the recommended A-YoloV5-based motorcycle detection model.

$$CM_{CCY} = \arg \min_{\{Y_c^{YOVS}, Ab_d^{YOVS}, O_i^{YOVS}\}} \left(\frac{1}{Bh_W + Y_{IU}} \right) \quad (2)$$

Here, the hidden neurons, number of epochs, and steps per epochs are defined using the terms Y_c^{YOVS} in the range of [5, 255], Ab_d^{YOVS} within the interval of [5, 50] and O_i^{YOVS} lies in between [300, 1000]. An accuracy and IoU are denoted by Bh_W and Y_{IU} , and it is mathematically provided in Eq. (3) and Eq. (4).

$$Bh_W = \frac{(XC + AQ)}{(XC + AQ + VN + GO)} \quad (3)$$

$$Y_{IU} = \frac{XC}{(XC + VN + GO)} \quad (4)$$

Term XC indicates the 'true positive', AQ defines the 'true negative', GO represents the 'false negative', and VN signifies the 'false positive'. The pictorial view of the proposed A-YoloV5-based motorcycle detection model is offered in Fig. 2. Then, the obtained detected motorcycles are indicated by Kl_M^{Det} .

Further, the detected motorcycles Kl_M^{Det} are inputted into the ROI extraction. ROI extraction is a process that identifies and isolates the unwanted portions of an image or video frame. The procedure involves assessing data to identify prospective areas based on colors, form, or additional visual

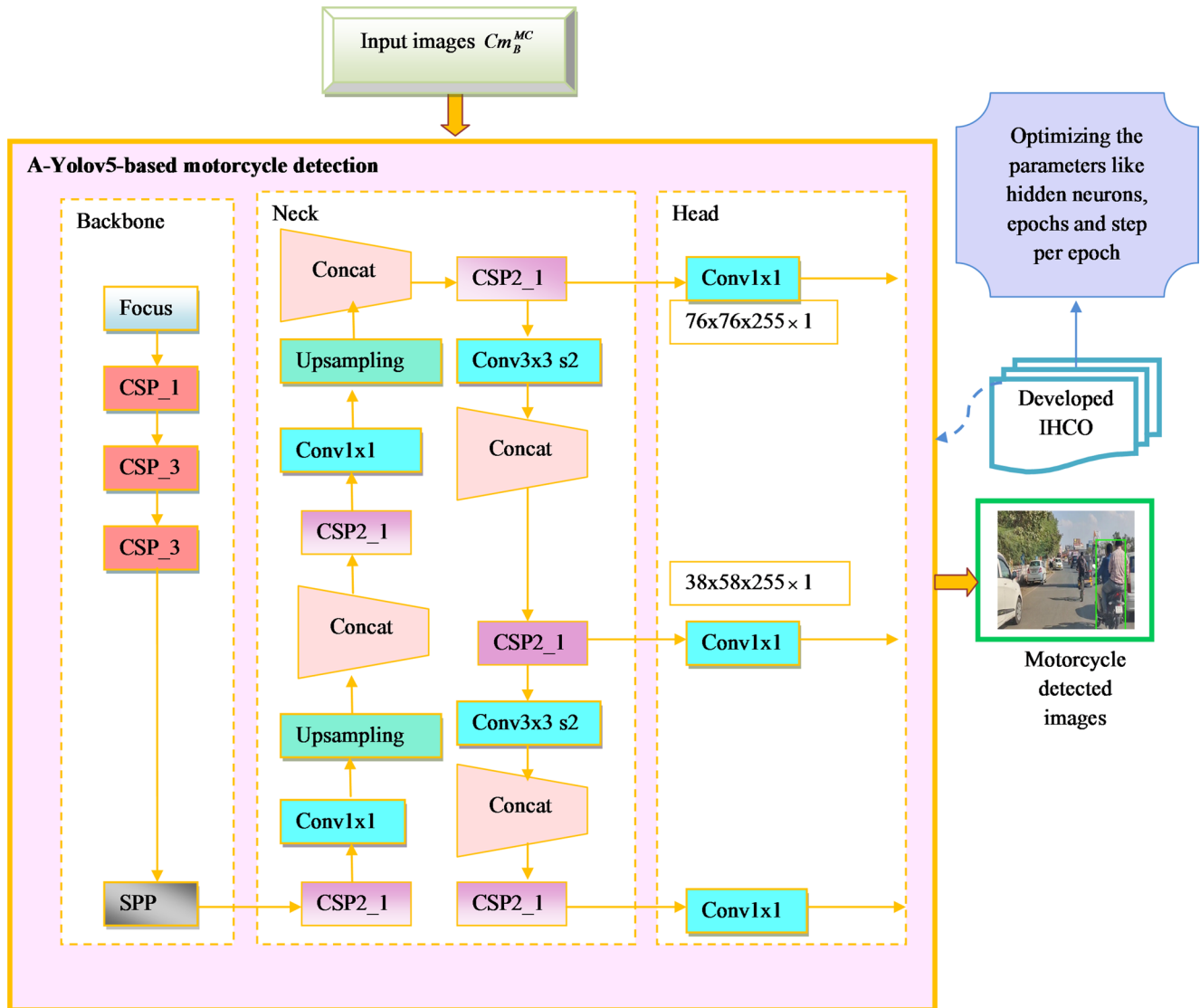


Fig. 2 Pictorial view of the proposed A-YoloV5-based motorcycle detection model

characteristics. Once possible regions have been discovered, the ROI extractor is used to extract them as distinct entities. The obtained ROIs extracted images are indicated by G_j^{Ext} and it is subsequently used for the helmet detection process. This ROI extraction aids in extracting the exact localization and evaluation of helmets in provided data, allowing for a variety of uses such as security tracking, sports evaluation, and industry inspection.

5 Helmet Detection and Number Plate Recognition for Secured Driving

This section describes the implementation and use of the proposed ViT-DRDNet for helmet detection and Adaptive Yolov5 for number plate recognition.

5.1 Vision Transformer

The Vision Transformer (ViT) (Chen 2021) splits the images into patched tokens, which are subsequently vertically displayed. A transformer encoding consists of segments, each including Multi-headed Self-Attention (MSA) and a Feed-Forward Network (FFN). FFN has two-layer Multilayer Perceptron (MLP) with an increasing proportion at the layer that is hidden, while a ‘Gaussian Error Linear Unit (GELU) operation is introduced following the initial regular layer. During each block, Layer Normalization (LN) is carried out, followed by residual savings. The ViT input B_0 to be processed within the p^{th} phase may be described using Eq. (5).

$$\begin{aligned} B_0 &= [B_{CS} || B_{PTH}] + B_{Ps} & (a) \\ C_p &= B_{p-1} + MSA(LN(B_{p-1})) & (b) \\ B_p &= B_p + FFN(LN(B_p)) & (c) \end{aligned} \quad (5)$$

Here, the classification [CLS] as well as patched tokens is represented by $B_{CS} \in V^{1 \times G}$ and $B_{PTH} \in V^{R \times G}$, accordingly, with the location embedded $B_{Ps} \in V^{(1 \times R) \times G}$.

5.2 Dilated Residual DenseNet

Dilated Residual DenseNet (DRD) (Yang 2020) is the major backbone. The innovative dilated component extracts data and combines them to create effective information for negative map estimates, building on previous improvements in dense as well as residual connections. DRD consists of four components: (a) the densely connected layer, (b) the transition layer, (c) the residual connected layer, and (d) the dilated module.

Dense Connected Layer A dense interconnected layer is created by sending the result of the previous dilated DenseNet and the outcome associated using Eq. (6) with the previous layer to the existing layer.

$$L_{Rs,n,m} = j_{Rs,n,m}([L_{Rs,0,m}, L_{Rs,1,m}, \dots, L_{Rs,n-1,m}]), \quad \text{if } 1 \leq n \leq H \quad (6)$$

Term $j_{Rs,n,m}$ signifies the function of the dilated component, $L_{Rs,n,m}$ represents the output characteristics of the n^{th} dilated component that comprises the m^{th} DRD, and $L_{Rs,m,0} = L_{Rs,m-1}$.

Transition Layer To take full advantage of the ordered characteristics obtained by every dilated component and automatically regulate the characteristics to the following unit, a layer of convolution using a kernel dimension was introduced and equivalent to 1×1 combination of every resultant data from the previous DRD and the entire sections in the present DRD and shown in Eq. (7).

$$L_{Tns,m} = j_{Tns,m}([L_{Rs,0,m}, L_{Rs,1,m}, \dots, L_{Rs,H,l}]) \quad (7)$$

Here, the term $[L_{Rs,0,m}, L_{Rs,1,m}, \dots, L_{Rs,H,l}]$ denotes the entire feature concatenation.

Residual Connected Layer The residual interconnection is used to maintain the movement of information from the lowest to the highest level, improving depiction capability through Eq. (8).

$$L_{Rs,m} = L_{Rs,m-1} + L_{Tns,m} \quad (8)$$

Dilated Aggregated Module The suggested DRD uses a dilated component to gather multi-scale data, unlike dense blocks or residual dense blocks. The dilated function mappings input features to a set number of outcome map features using convolutional having a kernel dimension of 1×1 .

5.3 Developed ViT-DRDNet for Helmet Detection

In this study, a ViT-DRDNet is developed to accurately detect helmets in video frames extracted from motorcycle detection outputs. The model leverages the attention mechanism of Vision Transformers to focus on relevant regions within the frame, while the Dilated Residual DenseNet backbone captures multi-scale spatial features, improves recognition of helmets even under challenging conditions. By combining these architectures, ViT-DRDNet provides high accuracy and robustness, making it well-suited for real-time helmet detection in traffic surveillance and construction site monitoring scenarios.

Novelty The developed ViT-DRDNet for helmet detection begins with the ROI extracted G_j^{Ext} from the motorcycle detection stage, ensuring that only the rider-specific region is analyzed when eliminating background interference. The extracted ROI is processed through the DRDNet backbone, which incorporates multiple dense blocks combined with dilated convolutional layers. The dense blocks promote feature reuse and efficient gradient propagation, enabling the network to capture fine-grained patterns of helmets such as shape, edge, and texture. The integration of dilated convolutions further enhances the receptive field without increasing computational complexity, allowing the model to effectively recognize helmets of different sizes, orientations, and partially occluded conditions. The features derived from these dense-dilated layers are then aggregated through additional convolutional layers to form compact multi-scale feature maps. In parallel, the ViT module processes the ROI by splitting it into fixed-size patches and converting them into embedded tokens, which are passed through a multi-head self-attention mechanism. This enables the network to selectively focus on the most informative regions of the ROI, emphasizing helmet-specific areas while suppressing irrelevant background information. The outputs of the DRDNet and ViT modules are fused together to create a rich feature representation H_P^{HDet} that leverages both local spatial details and global contextual awareness.

This hybrid design is the core novelty of model, as it synergizes CNN-based dense multi-scale learning with transformer-based attention mechanisms, leading to robust helmet detection even under challenging real-world conditions such as motion blur, illumination changes, and occlusions. Finally, the fused features are classified to determine the presence or absence of a helmet. If a helmet is detected, the result is to be stored, whereas in case of no helmet is identified, the framework automatically initiates the number plate recognition process using Adaptive YOLOv5, thus enabling automated enforcement and real-time alert generation. This dual-pathway approach not only improves accuracy and IoU performance but also makes the system highly reliable for intelligent traffic monitoring and safety applications. Figure 3 illustrates the graphical representation of the recommended ViT-DRDNet-based helmet detection model.

5.4 Developed A-Yolov5-based Number Plate Recognition for Message Alert

The number plate recognition process can be carried out using A-YoloV5. If the person does not wear a helmet on the specific motorcycle, then the developed A-YoloV5 model aids in recognizing the number plate. When the system identifies a number plate, it quickly sends a message alert to all

relevant authorities. This alert contains the vehicle's plate number, location, and the details of the infraction. The system is designed to detect number plates and helmets only, with no private data is captured or stored.

The ability to give real-time notifications in response to the identification of crucial safety occurrences like driving without a helmet is a major benefit of the alert mechanism in the developed model. This instant alert message to the registered phone number improves safe driving by quickly alerting the appropriate parties such as law enforcement or bike owners. It assists in addressing safety issues, enables prompt remedial measures to be implemented, such as stopping reckless driving or dealing with helmet regulation non-compliance. Through the facilitation of prompt intervention, the risk of accidents is highly minimized, and safer driving environments are created as this mechanism considerably enhances the overall security system.

6 Result and Discussion

The simulation findings of the proposed and detailed explanation regarding the obtained results are given in this section.

6.1 Experimental Setup

The suggested automated helmet recognition model was carried out by employing the Python platform. The introduced model utilized 10 populations, chromosome length was 3, and the highest iteration was 50. The segmentation models like UNET (Wang et al. 2023), TransUnet (Fang et al. 2022), MobileUnet (Pu et al. 2022) and the classifiers like CNN (Lee 2023), Resnet (Mathew 2023), DenseNet (Mugume et al. 2021), Residual Densenet (Res-Densenet) (Gong et al. 2022), R-CNN (Jia 2021), YOLOv3 (Kumar et al. 2024), YOLOv5 (Sarmah 2024) and DPPNet (Alassaf 2024), and the optimization approaches (Feng et al. 2024) like Arithmetic Optimization Algorithm (AOA) (Abualigah 2021), Black Widow Optimization (BWO) (Hayyolalam 2020), Spider Monkey Optimization (SMO) (Kumari et al. 2024) and Hermit Crab Optimizer (HCO) (Tafakkori and Tavakkoli-Moghaddam 2022) were used for comparison process (Nguyen et al. 2021) to showcase the effectiveness (Ji et al. 2023) of the suggested scheme.

6.2 Simulation Measures

The metrics used for evaluating the performance of proposed motorcycle and helmet detection mechanism are taken from the link (https://en.wikipedia.org/wiki/Precision_and_recall).

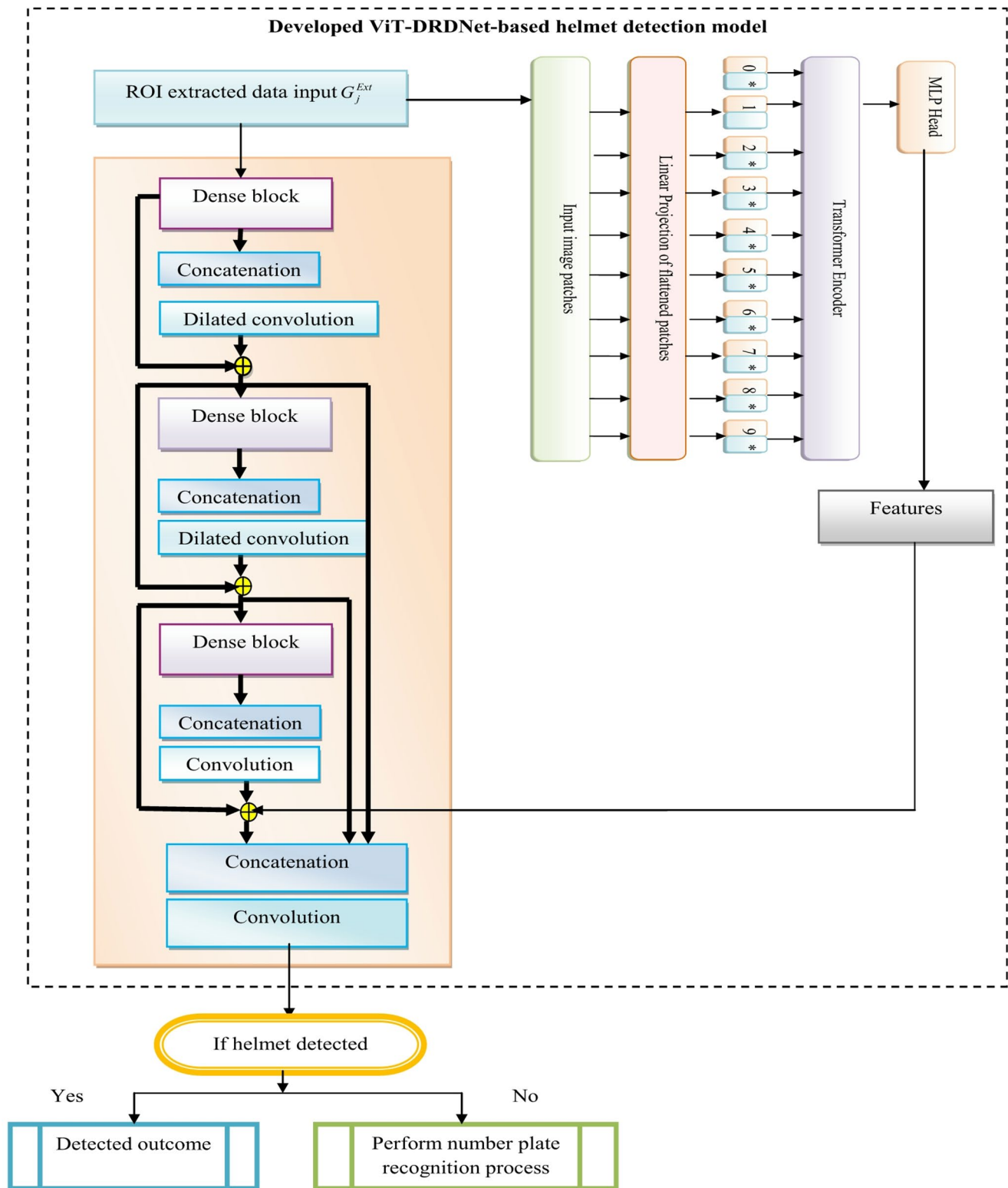


Fig. 3 Graphical representation of the recommended ViT-DRDNet-based helmet detection model

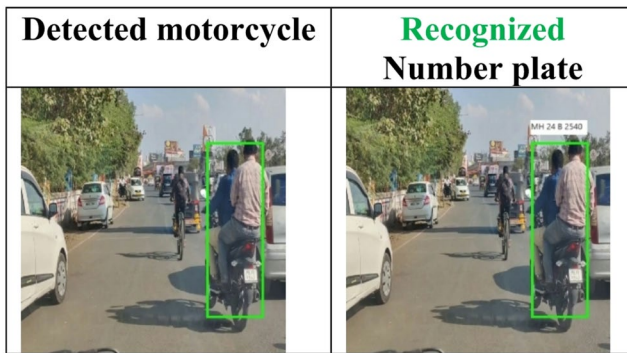


Fig. 4 Detected motorcycle and number plate results on the developed model

6.3 Detected Image Results

Figure 4 depicts the detected motorcycle and recognized number plate on the designed A-Yolov5 model.

6.4 Motorcycle Detection-aided Analysis

Figure 5 and Fig. 6 present the motorcycle detection-aided analysis of the developed model with traditional algorithms and detection techniques.

6.5 Helmet Detection-aided Comparison Analysis on the Developed Model

The designed ViT-DRDNet-based helmet detection comparison analysis on the developed model is given in Table 1.

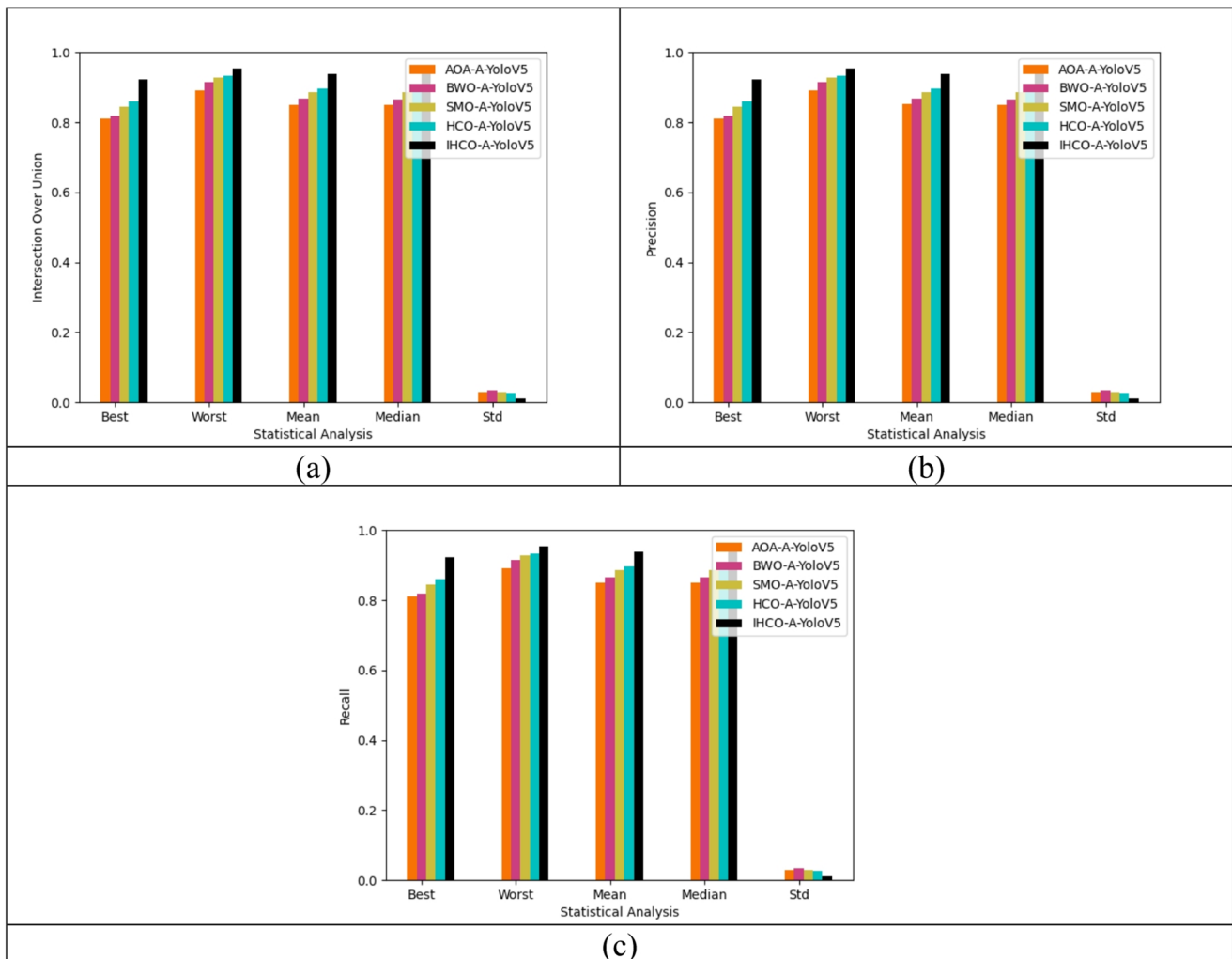


Fig. 5 Motorcycle Detection-aided Analysis on the Developed Model with Traditional Heuristic Approaches regarding (a) IoU, (b) Precision, and (c) Recall

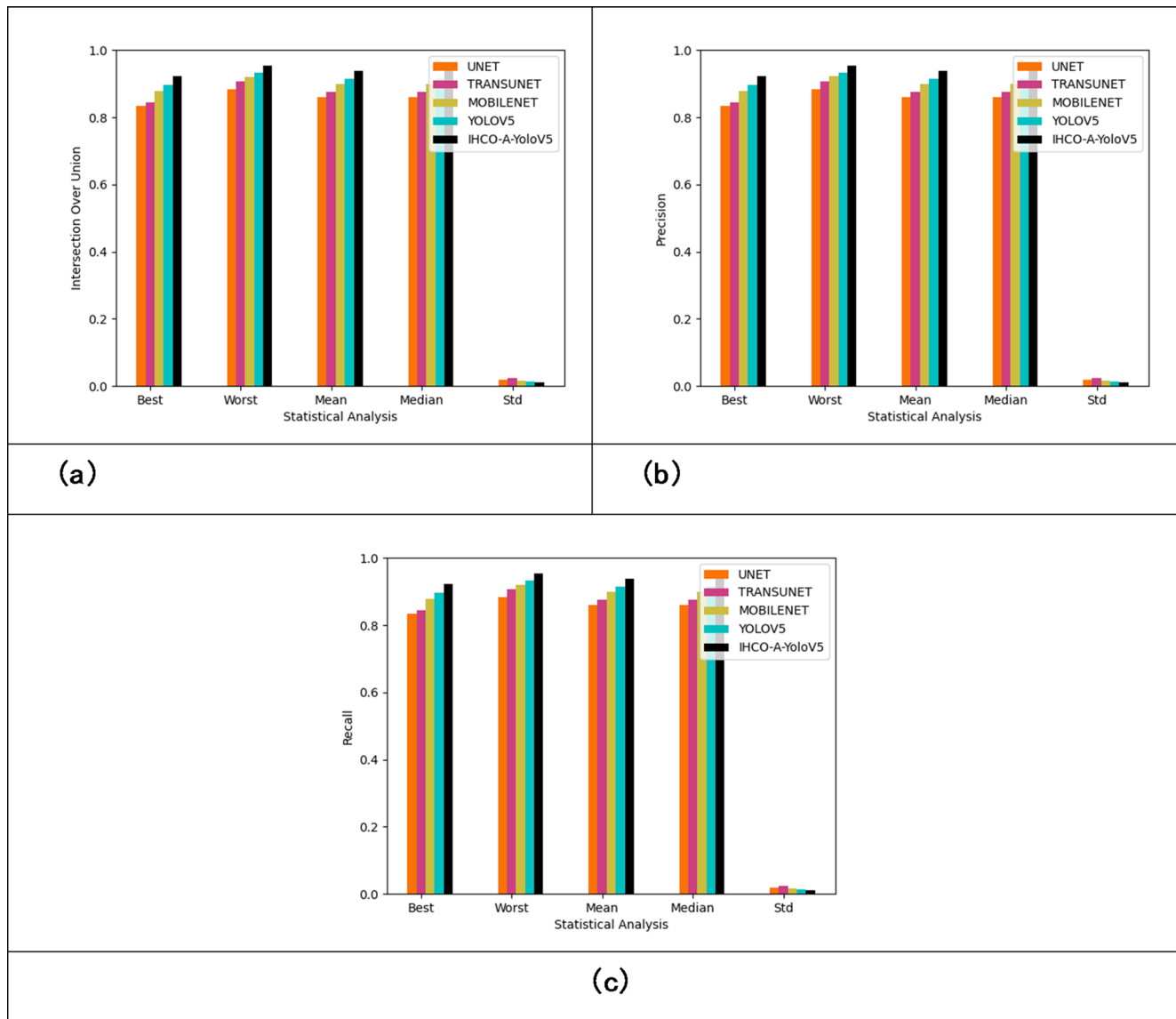


Fig. 6 Motorcycle Detection-aided Analysis on the Developed Model with Traditional Models regarding (a) IoU, (b) Precision, and (c) Recall

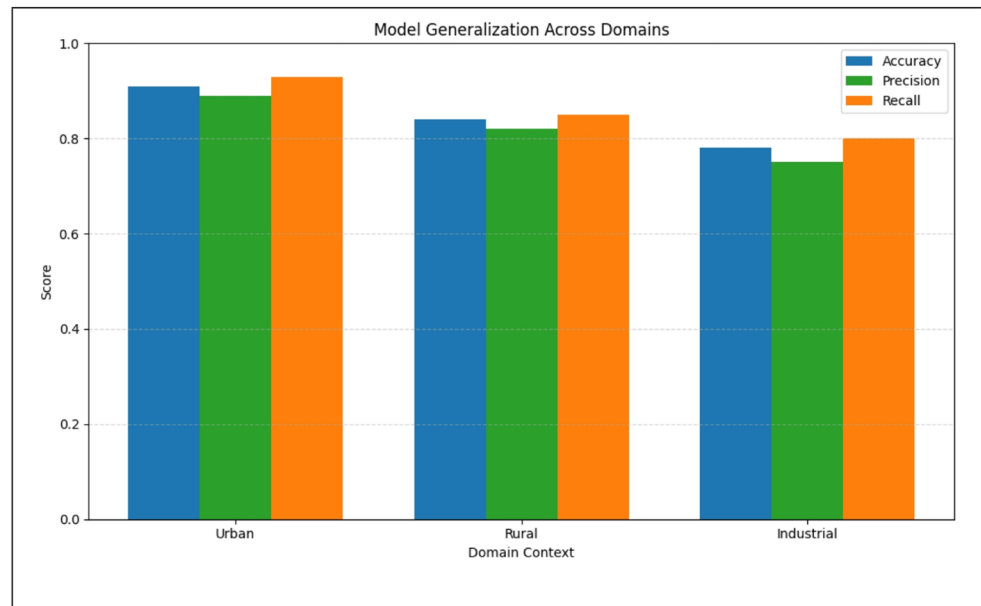
Table 1 Helmet Detection-aided comparison analysis on the developed model

Terms	R-CNN (Feng et al. 2024)	YOLOv3 (Kumar et al. 2024)	YOLOv5 (Sarmah 2024)	DPPNet (Alasaf 2024)	ViT- DRD- Net
Accuracy	89.004	91.173	89.819	93.834	94.952
Sensitivity	89.004	91.191	89.795	93.831	94.950
Specificity	89.005	91.155	89.844	93.837	94.955
Precision	89.005	91.158	89.839	93.837	94.955
FPR	10.995	8.845	10.156	6.163	5.045
FNR	10.996	8.809	10.205	6.169	5.050
NPV	89.005	91.155	89.844	93.837	94.955
FDR	10.995	8.842	10.161	6.163	5.045
F1-Score	89.004	91.175	89.817	93.834	94.952

6.6 Generalization Performance of the Proposed Model Using New Dataset

The developed model is further validated by newly included dataset covering diverse real-world traffic environments such as urban, rural, and industrial regions to prove the generalization ability of the developed model, which is demonstrated in Fig. 7. The evaluation results confirm that the developed ViT-DRDNet with Adaptive YOLOv5–IHCO framework consistently maintains high detection accuracy across these heterogeneous datasets, significantly outperforming conventional models. Thus, the strong performance across new datasets validates that the developed framework is robust, scalable, and suited for real-world helmet detection compared to prior approaches. The developed model

Fig. 7 Generalization performance of the developed model



consistently outperformed existing benchmark methods, achieving higher precision, recall, and F1-scores across all environments. Conclusively, the evaluation proves that the developed framework not only generalizes effectively across heterogeneous real-world scenarios but also demonstrates superior robustness and accuracy compared to traditional deep learning baselines, thereby validating its practical applicability for reliable helmet detection in real traffic environments.

6.7 Discussion

The results of this developed work demonstrate the efficacy of the suggested helmet identification system, with impressive performance measures across distinct evaluation criteria. Figures 5 and 6 show the motorcycle detection-aided analysis for the designed model over existing algorithms and techniques by employing measures such as IoU, precision and recall. The high IoU score represents the accurate localization of motorcycles and helmets, while the recall and precision values recommend a good balance among correctly recognizing the helmet motorcycles and reducing the false positives. The experimental results have shown that the designed IHCO-A-Yolov5 provides superior performance rate than the other models in the motorcycle identification process. Table 1 depicts the helmet detection-aided comparison analysis of the implemented approach over previous models. Here, the accuracy, sensitivity, specificity, precision, FPR, FNR, NPV, FDR, and F1 score measures are considered in this experiment. Here, the specificity, sensitivity, and accuracy reinforce the ability of the system to correctly classify the motorcycles without and with helmets. The less FNR and FPR values specify that the model

is robust and reduces the errors. The high F1 score, which balances the recall and precision, underscores the overall performance of the system. The NPV has additional insights into the reliability of the system in real-world scenarios. The performance of the system can be attributed to be effective utilization of ViT-DRDNet, which allows accurate detection. These results have significant implications for surveillance and road safety, as the system can be integrated with conventional infrastructure to enforce and monitor helmet utilization, highly minimizing the risk of fatalities and head injuries.

Limitation Despite the advancements of the developed model, it faces several limitations. However, an important limitation of the current framework lies in the process of accessing user details, which may pose potential security and privacy issues. Since sensitive information, such as vehicle registration details and user identity, is utilized for violation reporting, unauthorized access or misuse of such data could compromise user trust and system reliability.

Future Scope To address this issue, the future work will focus on integrating robust encryption techniques, multi-factor authentication mechanisms, and secure key management protocols to ensure that data remains protected throughout collection, transmission, and storage. By prioritizing privacy and security at each stage of pipeline, the framework can effectively mitigate risks associated with sensitive data handling, comply with legal and ethical requirements, and strengthen public confidence in the deployment of intelligent traffic monitoring systems. Finally, future research would emphasize the integration of verification mechanisms or manual override option to minimize false positives.

This will include multi-stage validation strategies, such as applying dynamic confidence thresholds and incorporating manual override options, like Human-In-The-Loop (HITL), where enforcement actions are reviewed or verified by authorized personnel before finalization, would enhance transparency and public trust. Exploring these directions will not only improve the legal defensibility of the system but also ensure fairness, accountability, and broader acceptance in real-world deployments.

6.8 Ethical and Privacy Considerations

In addition to technical performance, ethical aspects of deploying helmet detection and number plate recognition systems in real-world scenarios must be carefully considered. The deployment of automated surveillance systems integrated with number plate recognition raises critical ethical and legal concerns. Privacy implications arise from continuous monitoring of individuals and vehicles, which necessitates strict adherence to data protection regulations. To mitigate risks, the system must incorporate secure data storage, encryption of message alerts, and restricted access policies. Furthermore, transparency regarding data usage and retention is essential to maintain public trust. Legal compliance with regional and international surveillance and privacy standards should also be ensured before large-scale implementation. Addressing these aspects strengthens the reliability, acceptance, and defensibility of the proposed enforcement framework in real-world contexts.

6.9 Real-World Integration of the Developed Model

The proposed motorcycle and helmet detection system can be seamlessly integrated into existing traffic surveillance infrastructures with minimal additional setup. The primary requirement is the deployment of high-resolution CCTV or IP-based surveillance cameras at strategic traffic checkpoints, intersections, and highways, where helmet violations are frequent. These cameras continuously capture live video streams, which are then fed into the backend processing unit equipped with the Adaptive Yolov5 and ViT-DRDNet models optimized by the IHCO.

Once the motorcycle and helmet detection module identifies a violation, the number plate recognition system is activated. The recognized license plate data is then cross-verified with the national vehicle registration database to retrieve the registered owner's details. This information is automatically integrated with an alert system that generates a message notification. The alert is routed through a secured communication server and delivered directly to the violator's registered mobile number, informing them of the helmet violation and associated penalties.

From an enforcement perspective, the system provides a centralized monitoring dashboard accessible to traffic authorities. This dashboard displays real-time flagged violations, including detected images, license plate details, and timestamped video snippets, allowing officers to validate the evidence if needed. The inclusion of HITL review ensures that any ambiguous cases are manually verified before fines or legal actions are enforced, thereby minimizing false positives.

From a deployment perspective, the proposed framework is designed for seamless integration with existing video surveillance infrastructure. It can be connected to traffic monitoring cameras or security surveillance networks with minimal hardware modifications. The modular design ensures that the recognition pipeline, powered by IHCO, can be updated or scaled without disrupting the underlying surveillance systems.

In terms of the user interface, the developed system provides a dashboard that visualizes detected number plates, violation alerts, and recognition confidence levels in real time. This interface is intended for law enforcement operators, enabled it to make quick and reliable decisions. The interface also supports log management and retrieval functions, which is essential for evidence collection and auditing.

Finally, regarding scalability, the architecture of the proposed system allows deployment across multiple camera streams simultaneously, ensuring that performance is maintained in high-density urban environments. The lightweight optimization process minimizes computational overhead, making it as feasible to scale up to city-wide deployments without requiring prohibitively large computational resources. By contextualizing the deep learning with effective optimization algorithm within these practical dimensions, the contribution of the optimization technique is not only confined to algorithmic novelty but also extends to system-level improvements in accuracy, efficiency, and deployment readiness.

7 Conclusion

The developed helmet detection model was useful to assure the safety of the public by automatically identifying persons who didn't wear helmets. At first, the necessary videos were obtained from the relevant data resources. The motorbike was identified from the acquired video using the A-Yolov5 network. The network's efficiency was improved by optimizing the parameters of the network using the developed IHCO. The ROI was then retrieved from the observed video frames. Additionally, the ViT-DRDNet was developed to identify the helmet from the ROI retrieved frames. When

no helmet was spotted on a certain motorbike, the A-Yolov5 network identifies the motorcycle's number plate. Here also the comparable developed IHCO was used to optimize the network parameters. Depending on the vehicle's identified number plate, a message alert was sent to the authorized phone number. Finally, the efficacy and superiority of the suggested motorbike and helmet recognition mechanism were evaluated by comparing it to several current approaches and algorithms, highlighting the model's superiority, and the accuracy of the developed ViT-DRDNet-based helmet detection model is 6.68% more than CNN, 4.14% enhanced than ResNet, 5.71% effectual than DenseNet and 1.19% enhanced than Res-DenseNet. The result confirmed that the developed model could identify helmets in a variety of settings, such as outdoor areas, building sites, and athletic venues. Some of the applications of this work are parking lot security, industrial safety monitoring, traffic surveillance systems, and automated law enforcement.

Author contributions All authors have made substantial contributions to conception and design, revising the manuscript, and the final approval of the version to be published. Also, all authors agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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